

Organoid & Organ-on-Chip Industry Research

The Next Frontier in Human-Relevant Drug Discovery

Kaelyn Ka Ning Wen

Investment & Market landscape review
June 2026



Table of Contents

01

Market & Technology

- Executive summary
- Definition & scope
- Technology maturity
- Global Performance

02

Landscape & Drivers

- Geographic landscape
- Growth drivers
- Regulatory deep-dive
- Applications

03

Companies & Investment Thesis

- Emulate | Organoid Sciences | Daxiang
- Potential Opportunity in HK
- Asia: strengths & headwinds

Executive Summary

Human-relevant preclinical models are moving from research tools to regulatory-grade infrastructure

Organoids & Spheroids market size

Bn USD

2025

2034

CAGR

20.3%

~\$1.8B

~\$9.6B

Organ-on-chip market size

CAGR

29.7%

~\$201.5M

~\$2.1B

Revenue composition today, 2022, % of total

NA

48%

EU

34%

APAC

18%

Total = 100%

Revenue CAGR by region, 2019–2022, %

12%

4%

3%

APAC

NA

EU

APAC is growing ~3x faster than NA and EU, yet still represents the smallest share of revenue today — a scale-vs-growth gap.

3-5 yrs

FDA target to make animal testing the exception

Roadmap issued Apr 2025

Geography

The US currently owns the **revenue** base, APAC owns the **growth delta**. APAC remains the **smallest revenue** contributor today but is **growing faster** than North America and Europe, creating a scale-versus-growth mismatch.

Investment Angle

China and Korea are attractive entry points because policy support, lower experimental cost, large patient access, and faster clinical validation can compress the platform-validation cycle.

Recommendation

Back APAC platform leaders with validated disease models, standardized assay workflows, pharma partnerships, and a credible path from CRO revenue to scalable platform economics.

Definition & Scope

Organoids, Organ-on-chip, microphysiological systems, and complex in vitro model/CIVM

ORGANOIDS

Self-organizing 3D cell clusters grown from stem or primary cells that mimic an organ's architecture.

来源	多能干细胞 (Pluripotent stem cells, PSCs)		成体干细胞 (Adult stem cells, ASCs)	
	胚胎干细胞 (Embryonic stem cells, ESCs)	诱导多能干细胞 (Induced pluripotent stem cells, iPSCs)	健康组织来源的类器官	肿瘤类器官 (Patient-derived organoids, PDOs)
种类	大脑类器官、肝类器官、肾类器官等		包括几乎所有消化系统的类器官(肝脏、胰腺、结直肠、胃等), 部分非消化系统的类器官, 如呼吸系统(肺泡、肺支气管), 生殖系统(卵巢、宫颈、子宫内膜、阴道上皮等)	乳腺癌、肺癌、前列腺癌、食管癌、胰腺癌、膀胱癌、肠癌、胃癌等
特点	具有无限自我更新并分化为几乎所有器官特异性的细胞类型; ESCs来源因其应用涉及伦理问题导致使用受限		通常来源于患者的活检组织, 可分化成与组织高度一致的各种细胞类型, 具有自组装能力, 组织结构与细胞类型高度仿生; 一般仅含器官上皮部分, 缺乏基质、神经和血管系统	可保留肿瘤组织的生物学特征和异质性, 具有多次传代后基因组保持稳定、培养周期短等优势

- Structural & genetic fidelity
- Patient-derived; ideal for oncology
- Static culture — limited flow & mechanics

Table 1. The classification of organoids and their characteristics

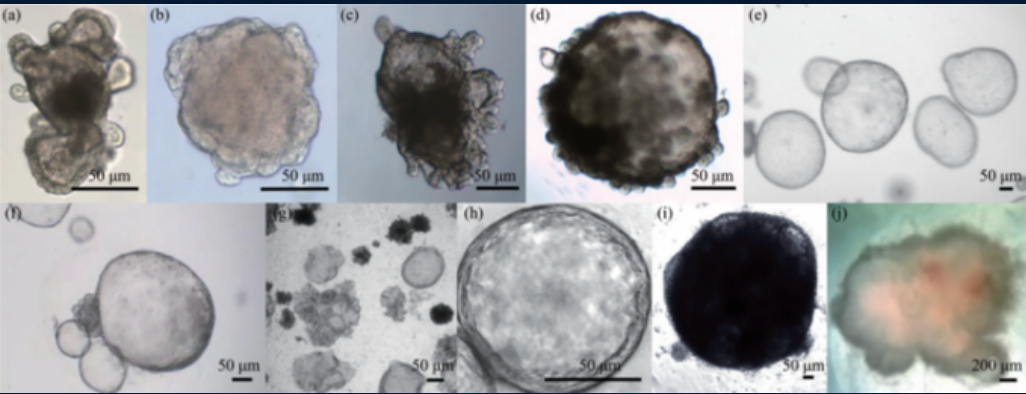
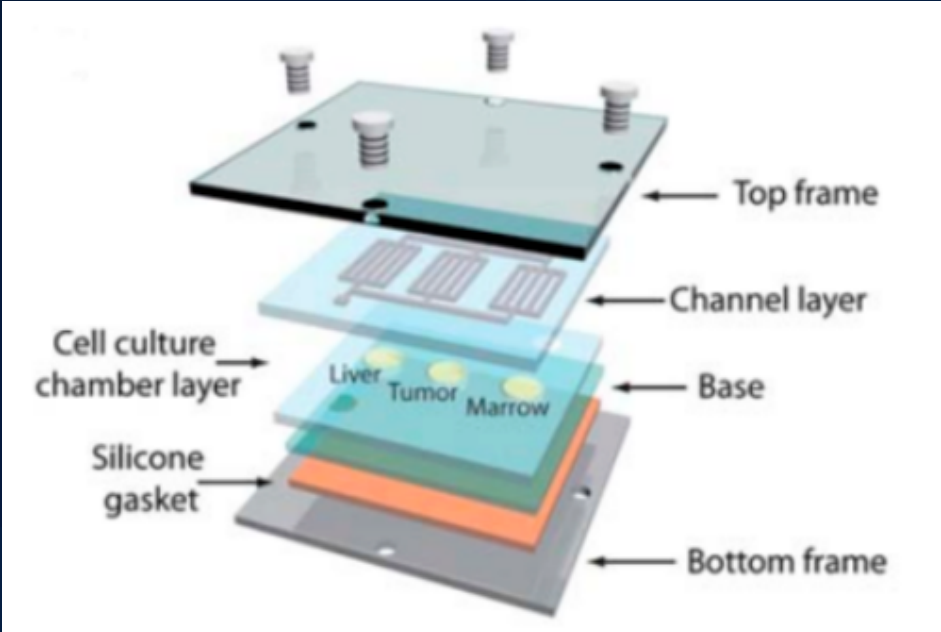


Table 2. (a) Small intestine organoid topography. (b) Colon organoid topography. (c) Esophageal organoid topography. (d) Stomach organoid topography. (e) Liver organoid topography. (f) Pancreatic organoid topography. (g) Lung organoid topography. (h) Breast organoid topography. (i) Renal organoid topography. (j) Brain organoid topography.

ORGAN-ON-CHIP

Microfluidic devices lining living cells in channels with flow, shear and mechanical cues.

It is called a "chip" because the microfabrication method it initially adopted was improved from the manufacturing method of computer microchips.



- Dynamic, physiological micro-environment
- Models organ-level function & crosstalk
- Often uses cell lines, not full 3D tissue

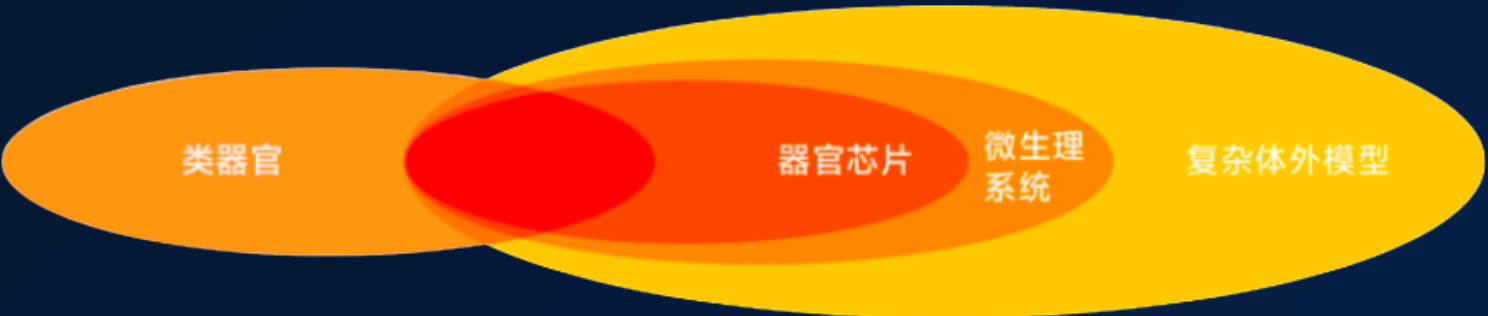
Table 3. Multi-organ microfluidic frame

Microphysiological system

A human cell-based in vitro platform that recreates key tissue or organ functions using 3D culture, microfluidics, biomaterials, and controlled cues such as flow, oxygen, pH, and mechanical stress.

Complex In Vitro Model / CIVM

Broader category of advanced biological models, including organoids, spheroids, 3D co-cultures, 3D bioprinted tissues, organ-on-chip systems, and MPS platforms.



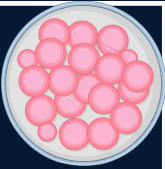
Technology maturity

Moving from proof-of-concept to a scalable, regulator-grade platform

Increasing biomimicry, throughput & regulatory readiness →

01

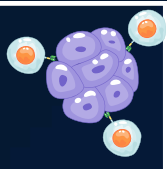
2D cell culture



Legacy baseline — cheap and simple, but poor human relevance.

02

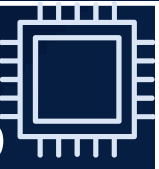
Organoids & spheroids



Self-organizing 3D tissue — real architecture, but static and avascular.

03

Organ-on-chip



Perfusion adds flow, shear & mechanical cues — but often 2D cell lines.

04

Organoid-in-tube



Patient-derived organoids integrated with microfluidic perfusion to model human tissue response under dynamic, clinically relevant conditions.

Evaluation Metric	2D Cell Model	Simple Animal Model	Non-human Primate	PDX	3D Organoid	Single Organ-on-Chip	Multi-Organ-on-Chip
Difficulty of model construction	Simplest	Moderate	Very difficult	Difficult	Moderate	Difficult	Very difficult
Modeling cycle	Short	Relatively short	Longest	Very long	Moderate	Long	Very long
Construction cost	Lowest cost	Relatively low cost	Highest cost	Very high cost	Relatively low cost	Moderate cost	Very high cost
Construction success rate	Highest	High	Moderate	Relatively low	High	Moderate	Low
Clinical relevance	Relatively low	Relatively low	Highest	Relatively high	High	High	High
Difficulty of in vivo observation	Low	Difficult to conduct	Difficult to conduct	Difficult to conduct	Relatively low	Relatively low	Relatively low

Applications

Drug toxicity & safety

- Predict liver, kidney & cardiac toxicity earlier — the largest current revenue pool.

Disease modeling

- Recreate cancer, NASH, infectious & rare diseases with human-relevant fidelity.

Personalized / precision oncology

- Patient-derived organoids guide targeted therapy selection and dosing.

Drug discovery & screening

- Higher-throughput platforms (e.g. 96 parallel chips) compress preclinical cycles.

Regenerative medicine

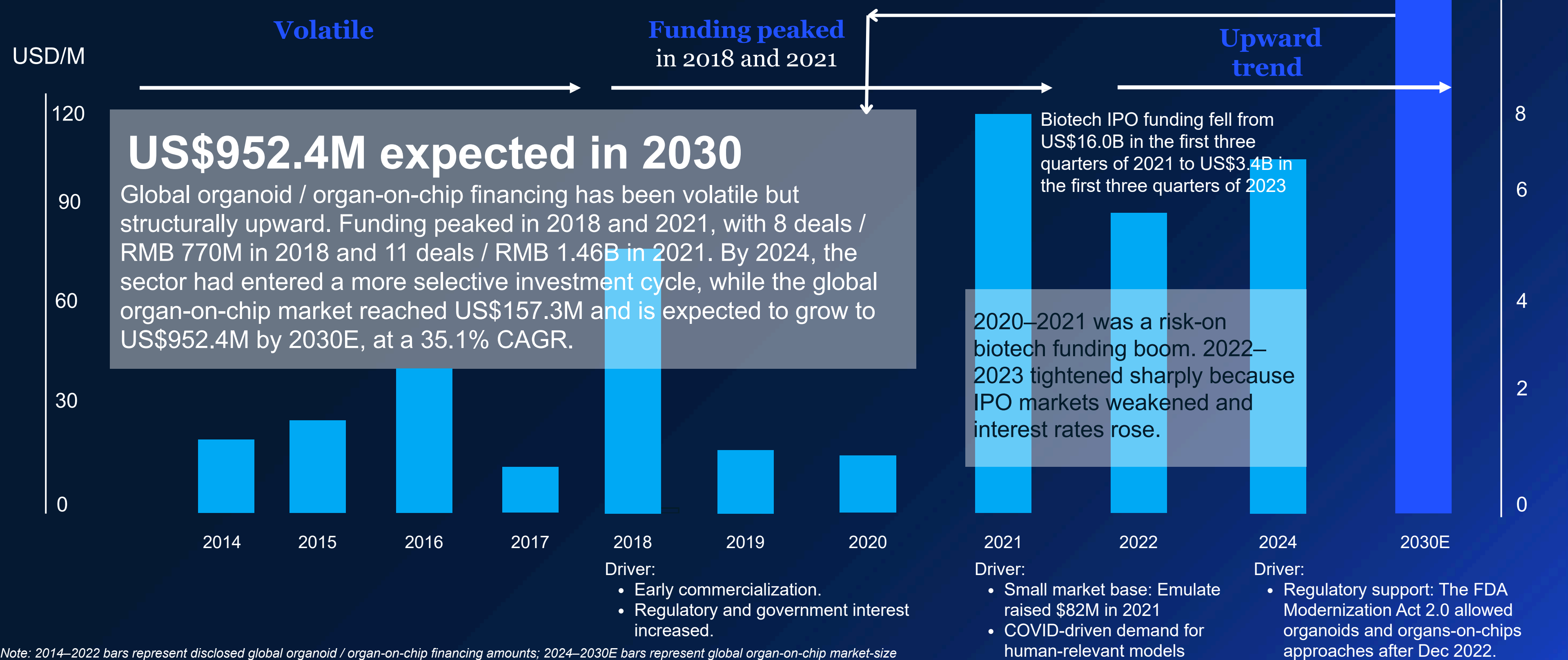
- Organoid transplantation & tissue engineering — fastest-growing application segment.

Sources: Eggshell Research Institute / VBData 2023; Sino Biological; Li et al., Protein & Cell, 2025; Zushin et al., JCI, 2023; DeepTech 2023; eMedClub 2023–24; Elveflow 2024.

Global Performance

Early-Stage Funding Volatility, Long-Term Platform Growth

A review of global organoid / organ-on-chip financing and market growth, 2014–2030E



Note: 2014–2022 bars represent disclosed global organoid / organ-on-chip financing amounts; 2024–2030E bars represent global organ-on-chip market-size estimates, not financing. Sources: VBData / VCBeat & Eggshell Research Institute 2023; Grand View Research 2025; McKinsey 2023; FDA / JCI 2023.

Table of Contents

01

Market & Technology

- Executive summary
- Definition & scope
- Technology maturity
- Market size & growth

02

Landscape & Drivers

- Geographic landscape
- Growth drivers
- Regulatory deep-dive
- Applications

03

Companies & Investment Thesis

- Emulate | Organoid Sciences | Daxiang
- Potential Opportunity in HK
- Asia: strengths & headwinds

Geographic landscape				
Global organoid & organ-on-a-chip (OOC) landscape by business model				
Players span the value chain — from platform developers to CROs, pharma-integrated units and instrument/reagent suppliers				
Company	HQ	Technology focus	Moat	Stage
Pure-play platform / technology developers				
Emulate	US — Boston	Organ-Chip systems (lung, liver, gut, brain)	OOC pioneer; automated "Zoë" instrumentation	Series E (\$225M+ total)
Mimetas	Netherlands	OrganoPlate microfluidic 3D-tissue plates	High-throughput	Series B (\$20.5M total)
Hesperos	US — Florida	“Human-on-a-chip” multi-organ systems	Functional, label-free readouts	Early Stage (by NIH/SBIR)
CRO / service providers (drug-discovery testing)				
Crown Bioscience	US / China	Patient-derived organoid (PDO) screening	Massive oncology tissue bank	Acquired by ADICON (\$204M)
Charles River	US	Integrated organoid & in-vitro CRO services	Global preclinical CRO leader	Public (CRL), Market Cap:~\$9.5B
Cellesce	UK	Organoid bioprocessing / scale-up	Reproducible automated manufacturing	Acquired by Danaher
Reagent, instrument & consumable suppliers				
STEMCELL Tech.	Canada	Organoid culture media & reagents	Category-standard media base	Private, partnership-backed
Corning	US	3D culture plates, Matrigel matrices	Ubiquitous global distribution	Public (GLW), Market Cap: ~\$154B+
CN Bio	UK	PhysioMimix OOC instruments	Strong FDA collaboration on safety trails	Series B (~\$21M first close)
Pharma-integrated / in-house platforms				
Roche / Genentech	CH / US	Internal organoid & OOC discovery units		
AstraZeneca	UK	In-house OOC for tox & efficacy		

Demand Drivers

Four forces are compounding demand for human-relevant models

Policy is the catalyst: a global shift away from animal testing

Dec 2022 FDA Modernization Act 2.0
Removes the decades-old mandate that drugs be animal-tested before human trials, explicitly permits organ-on-chips & organoids.

Apr 2025 FDA Roadmap + NIH mandate
FDA targets making animal testing the exception within 3–5 years. NIH requires human-relevant NAMs in new funding notices.

Dec 2025 FDA Modernization Act 3.0 (Senate)
Passed the Senate. Directs FDA to build a formal qualification pathway for non-animal methods and awaits the House.

2021–25 Asia & Europe align
China's 14th Five-Year Plan prioritizes stem-cell / tissue engineering (NMPA: 228 novel drugs in 2024). the UK roadmap (Nov 2025). Korea MFDS.

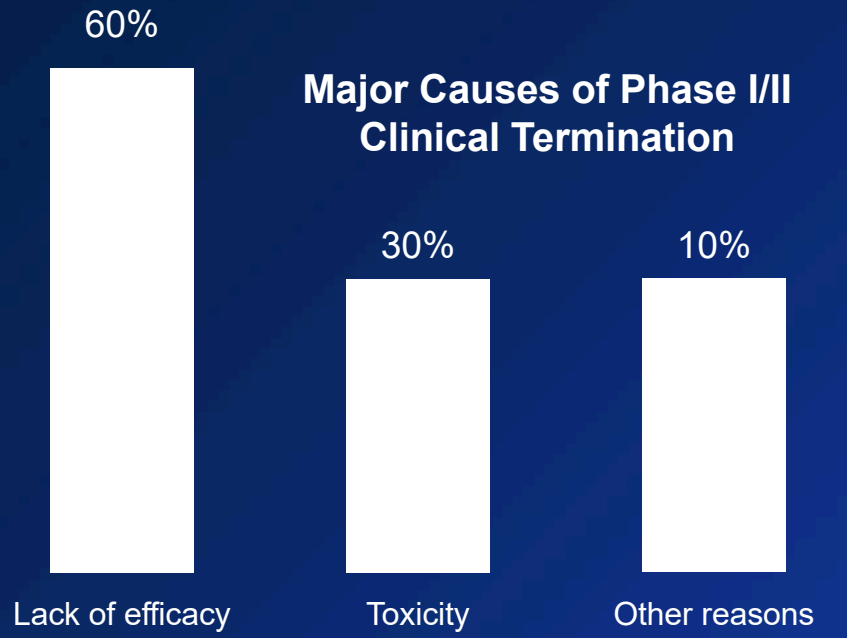
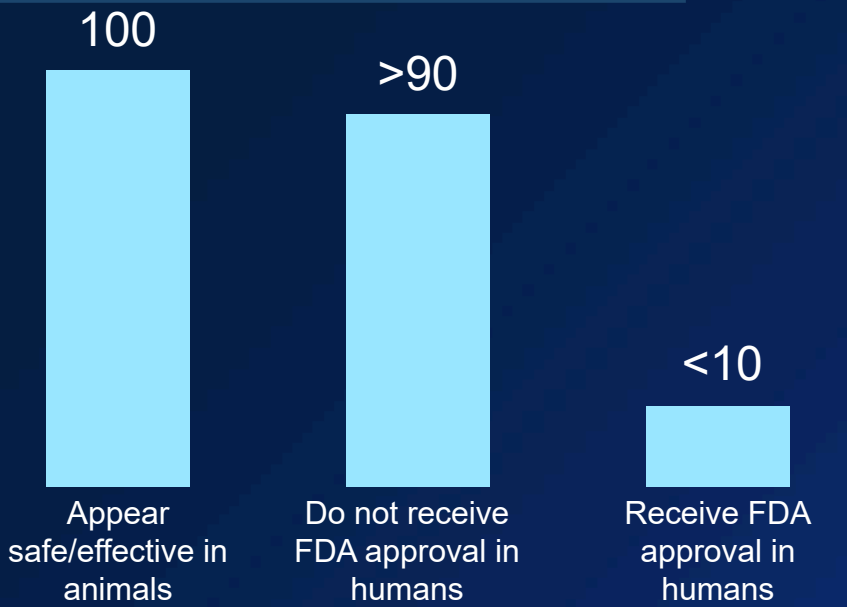
FDA Achieves Year 1 Goals in Reducing Animal Testing in Drug Development

Regulatory tailwind
The FDA, NIH, China, and the UK are explicitly steering preclinical testing toward non-animal new approach methodologies.

Precision & personalized medicine
Patient-derived organoids provide individual drug-response prediction, especially in oncology.

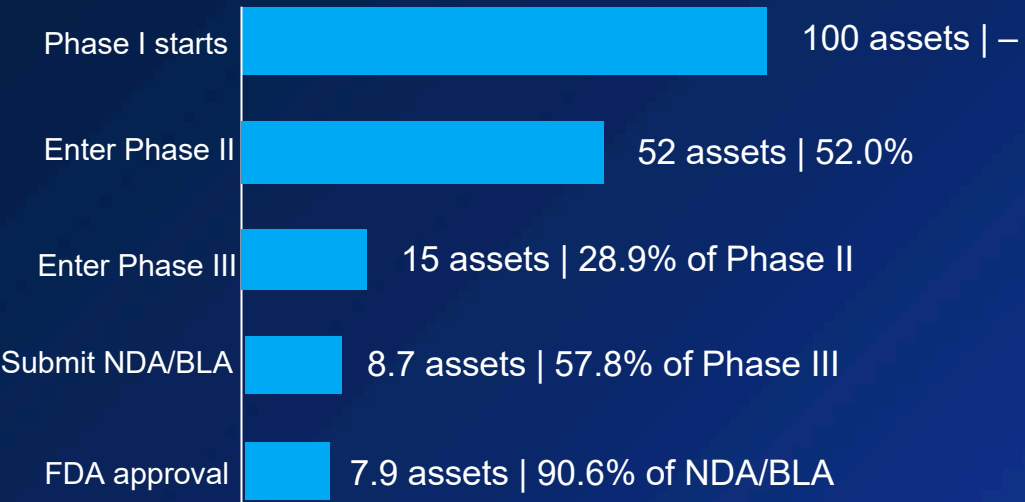
Predictive failure of animal models
~90% of drugs that clear animal testing still fail in humans. Human-cell models could provide better translation.

Pharma cost & speed pressure
~\$2B and 10+ years per approved drug drives adoption of faster, cheaper human-relevant screening.



Clinical Development Attrition Funnel Indexed to 100 Phase I Assets

Only ~7.9 of 100 Phase I assets reach FDA approval.



Kaelyn Ka Ning Wen

Sources: FDA; NIH; U.S. Congress; UK DSIT; DeepTech 2023; Protein & Cell 2025; JCI 2023; eMedClub 2023–24; BIO / QLS 2021.

Table of Contents

01

Market & Technology

- Executive summary
- Definition & scope
- Technology maturity
- Market size & growth

02

Landscape & Drivers

- Geographic landscape
- Growth drivers
- Regulatory deep-dive
- Applications

03

Companies & Investment Thesis

- Emulate | Organoid Sciences | Daxiang
- Potential Opportunity in HK
- Asia: strengths & headwinds

Company Profile • United States

Emulate — the category-defining incumbent



Founded

2014, spun out of Harvard's Wyss Institute

Funding

~\$225M raised; \$82M Series E led by Northpond

Core Moat

validated Liver-Chip S1 + AVA high-throughput system (96 independent Organ-Chip samples/run)

Footprint

150+ labs, 17 of top 25 biopharma

Edge

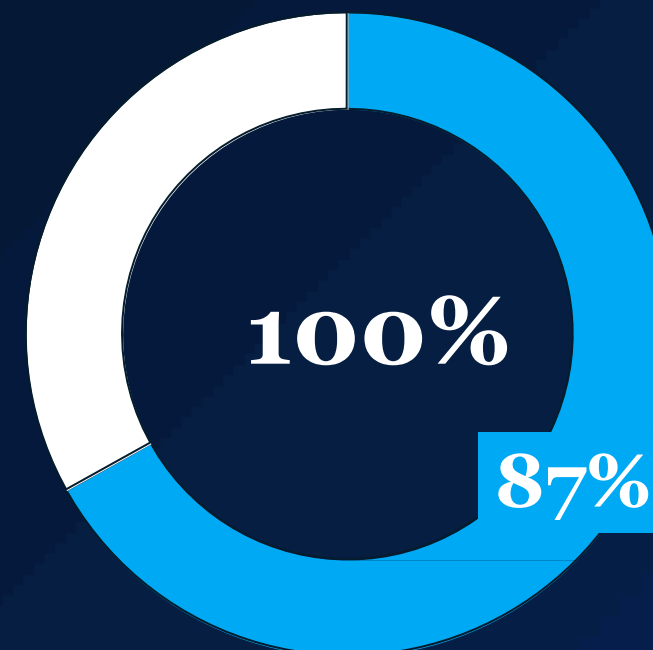
- First organ-chip accepted into FDA ISTAND for DILI prediction
- NASA AVATAR space-biology validation use case

Platform Expansion

liver, kidney, intestine, brain, AAV optimisation, disease modelling, and space-biology use cases

Emulate commercialized the Wyss Institute's organ-chip and remains the reference platform for the industry. Its peer-reviewed Liver-Chip data gave regulators the evidence base behind the FDA Modernization Act — making the company both a technology and a policy leader.

Liver-Chip S1 sensitivity / specificity for drug-induced liver injury (DILI)



100%
Specificity

Outperforming animal models in the largest head-to-head study of its kind.

Next-generation product

Liver-Chip R1 uses a rigid plastic chip to reduce small-molecule absorption, addressing the PDMS limitation in OOC systems.

Company Profile • Korea

Organoid Sciences — commercial-stage Korean scale-up



Core Moat

ODISEI — commercial since 2021

Revenue

~KRW 4B (2025), targeting ~KRW 30B by 2028

Focus

Organoid drug screening + next-gen cell therapies

Regulatory fit

Aligned to Korea MFDS & FDA Modernization Act

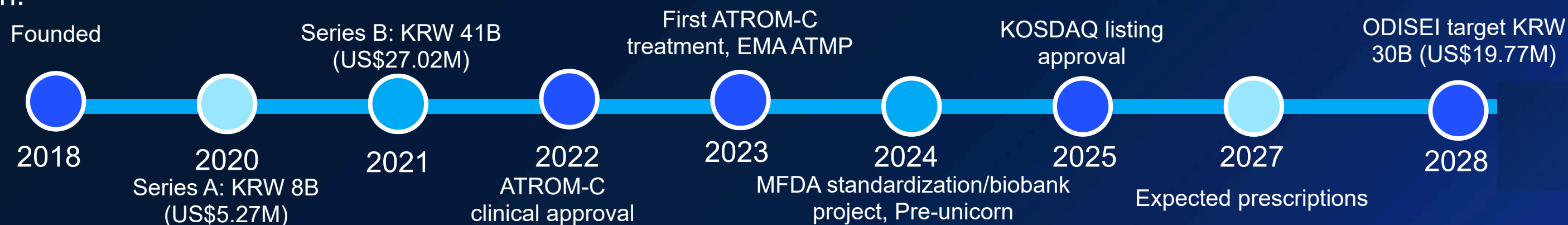
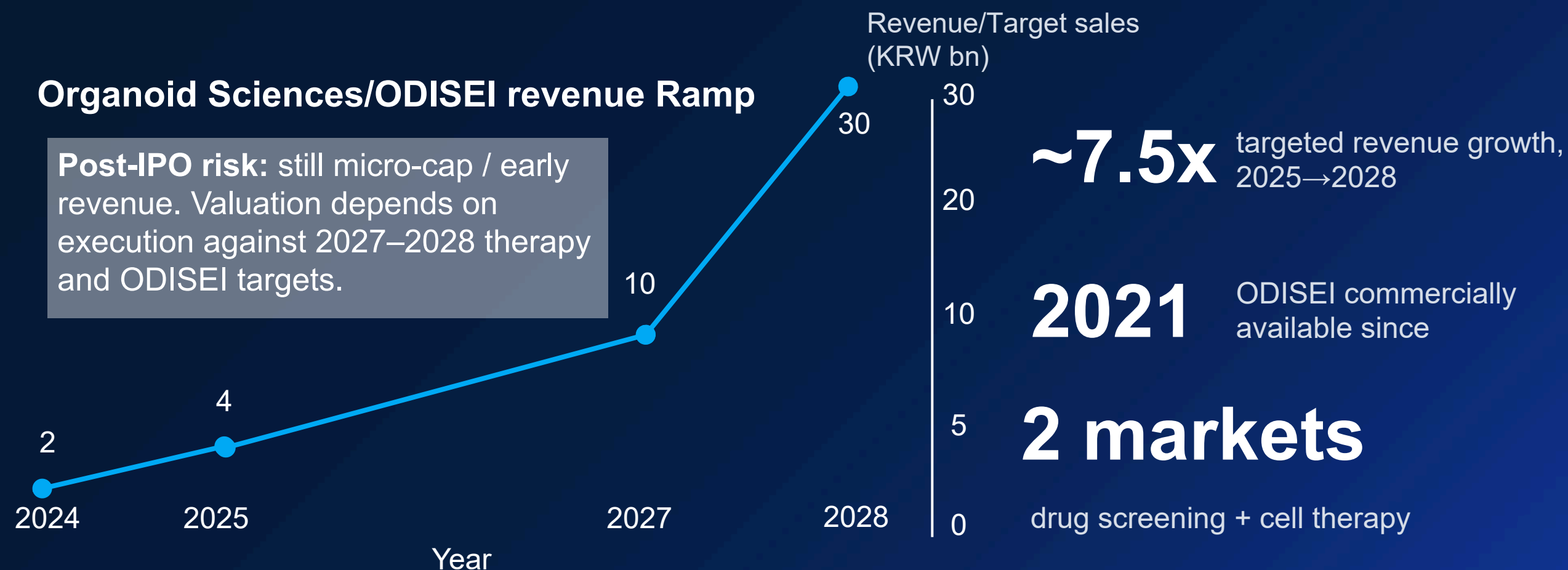
Status

- Revenue-generating platform with 2028 target
- Still needs proof of repeatable enterprise adoption.

Organoid Sciences is one of the few Asian players already generating commercial revenue. A credible 7-8x revenue ramp to 2028 signals real market pull. Dual positioning across drug-screening tools and cell therapy gives it two shots on goal, and explicit alignment with both Korean and US regulatory frameworks lowers cross-border adoption risk for a global investor.

Organoid Sciences/ODISEI revenue Ramp

Post-IPO risk: still micro-cap / early revenue. Valuation depends on execution against 2027–2028 therapy and ODISEI targets.



Company Profile • China



Beijing Daxiang — China's organoid-on-chip company (2018)

Funding

Series A led by CDH VGC; A+ round followed, with two rounds totaling around RMB100M.

Core Platform

IBAC pump-free, gravity-driven perfusion (ROCKER) — fuses organoid + organ-chip

Product breadth

Company-disclosed 4 self-owned IBAC® organoid-chip

Partners

Nikon alliance for organoid-chip imaging and translational application.

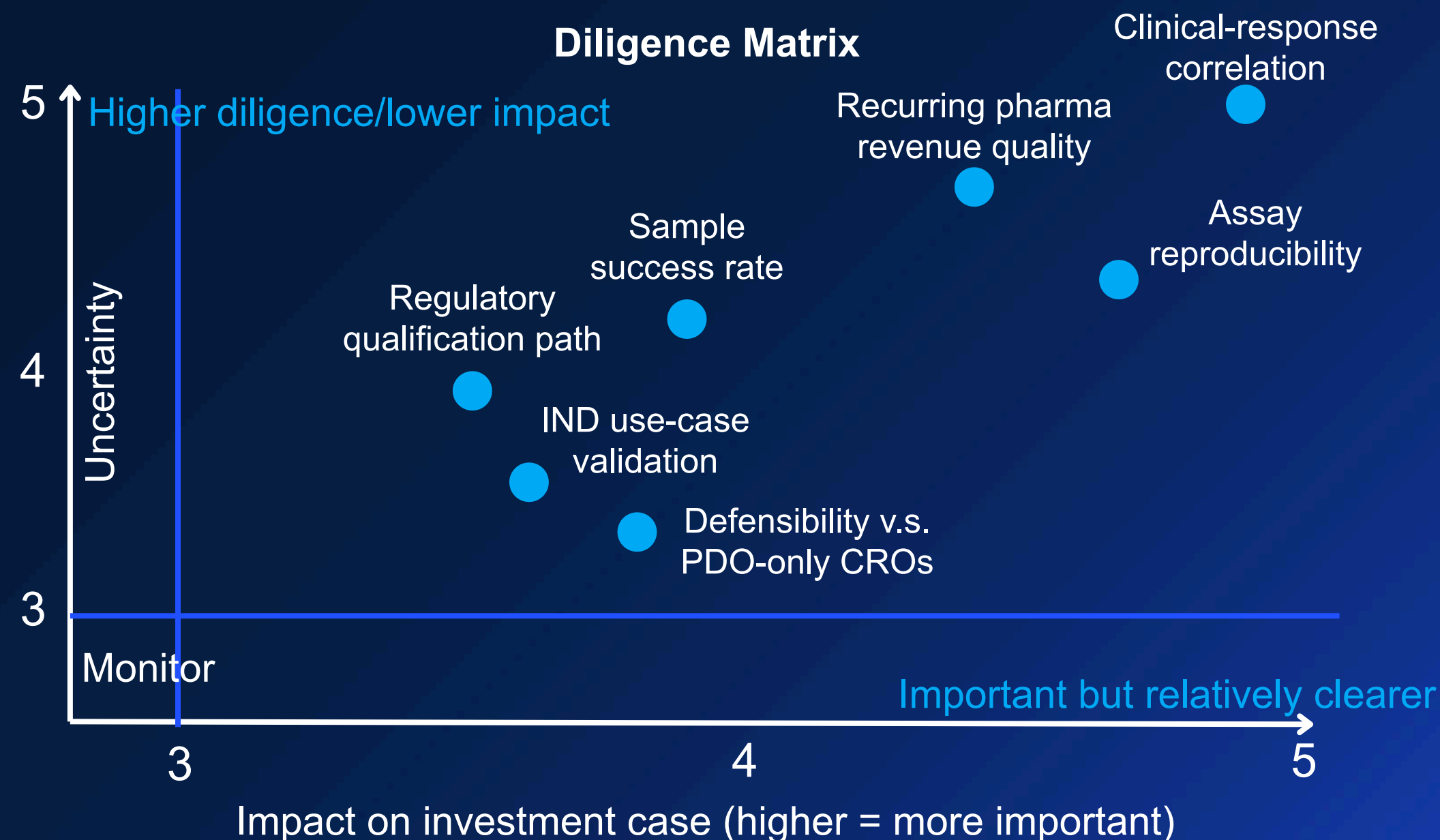
Traction signals

- 3 commercial organ-chip products launched in 2020
- 40+ China patents + 3 PCT patent items
- ISO9001 quality-system certification

concern

Verify hospital-network depth, assay reproducibility, sample success rate, IND use cases, and repeat pharma revenue

Daxiang is among the first Chinese players to commercialize the higher-fidelity organoid-on-chip approach, not just standalone organoids or chips. Its pump-free design lowers cost and complexity — a structural fit for high-throughput, price-sensitive Asian markets. Deep hospital partnerships give it the clinical data flywheel that Western incumbents guard closely, and it is riding China's explicit national policy support for the field.



Potential Opportunity · Hong Kong

HKU Organ-in-Cube / Liver-in-Cube, Prof. Kwan MAN's Team



Large unmet need

Liver cancer is the 6th most common cancer globally and the 3rd leading cause of cancer-related death

Moat

World's first 3D-bioprinted platform that reconstructs a patient-specific liver cancer microenvironment using the patient's own cells and extracellular matrix.

Differentiation vs standard organoids

It combines normal tissue, tumour tissue, vascular structures, immune microenvironment, tissue stiffness, and AI-assisted patient-specific printing, making it more biomimetic than conventional 3D culture or organoid models.

Commercialisation signal

The platform has started patient recruitment at local hospitals and is undergoing clinical trials in public and private hospitals, with HKU aiming to commercialise locally and overseas.

Validation

- Gold Medal at the Geneva International Exhibition of Inventions
- Special Grand Prize from the China Invention Association
- Best-Performing Start-Up Award at Asia Summit on Global Health.

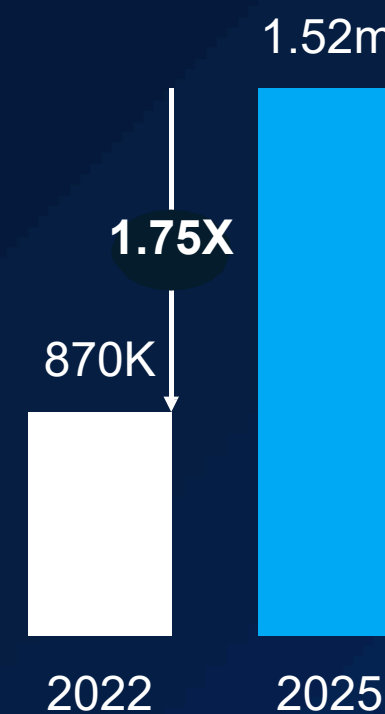
strong spin-off candidate

With Gobi–HKU Fund I now targeting HKU spin-offs, this platform could be commercialised as a precision oncology and pharma-testing company, starting from liver cancer and expanding into broader solid-tumor drug screening.

Expansion potential

Technology framework can expand beyond liver cancer into other liver diseases and solid tumours, creating a platform opportunity rather than a single-indication tool.

Liver Cancer New Cases Annually Projection



The Asia Opportunity

Strong tailwinds, real friction – baked by deep government support

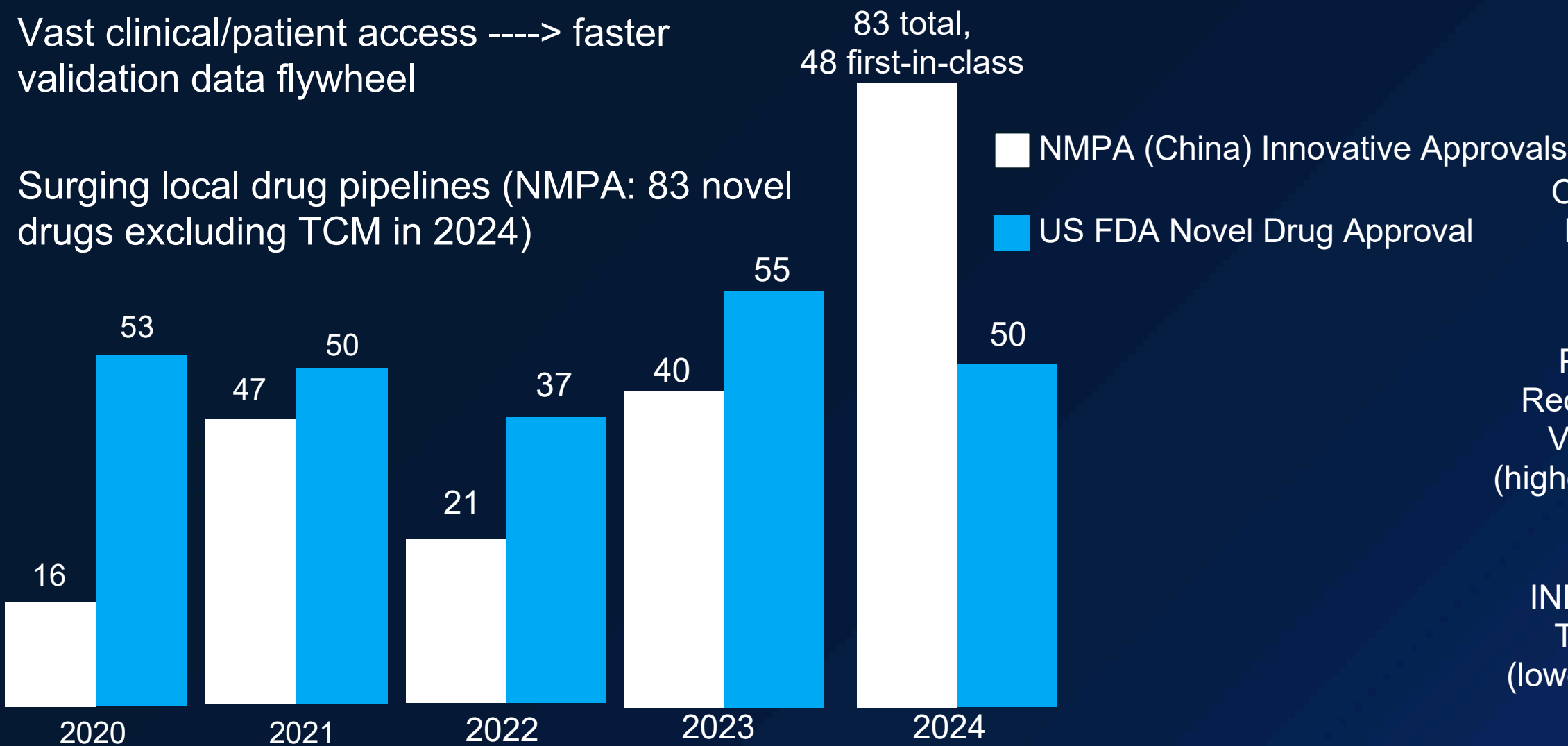
Strengths

Explicit national policy: China’s 14th FYP names stem-cell & tissue engineering priorities

Cost advantage in manufacturing and high-throughput screening

Vast clinical/patient access ----> faster validation data flywheel

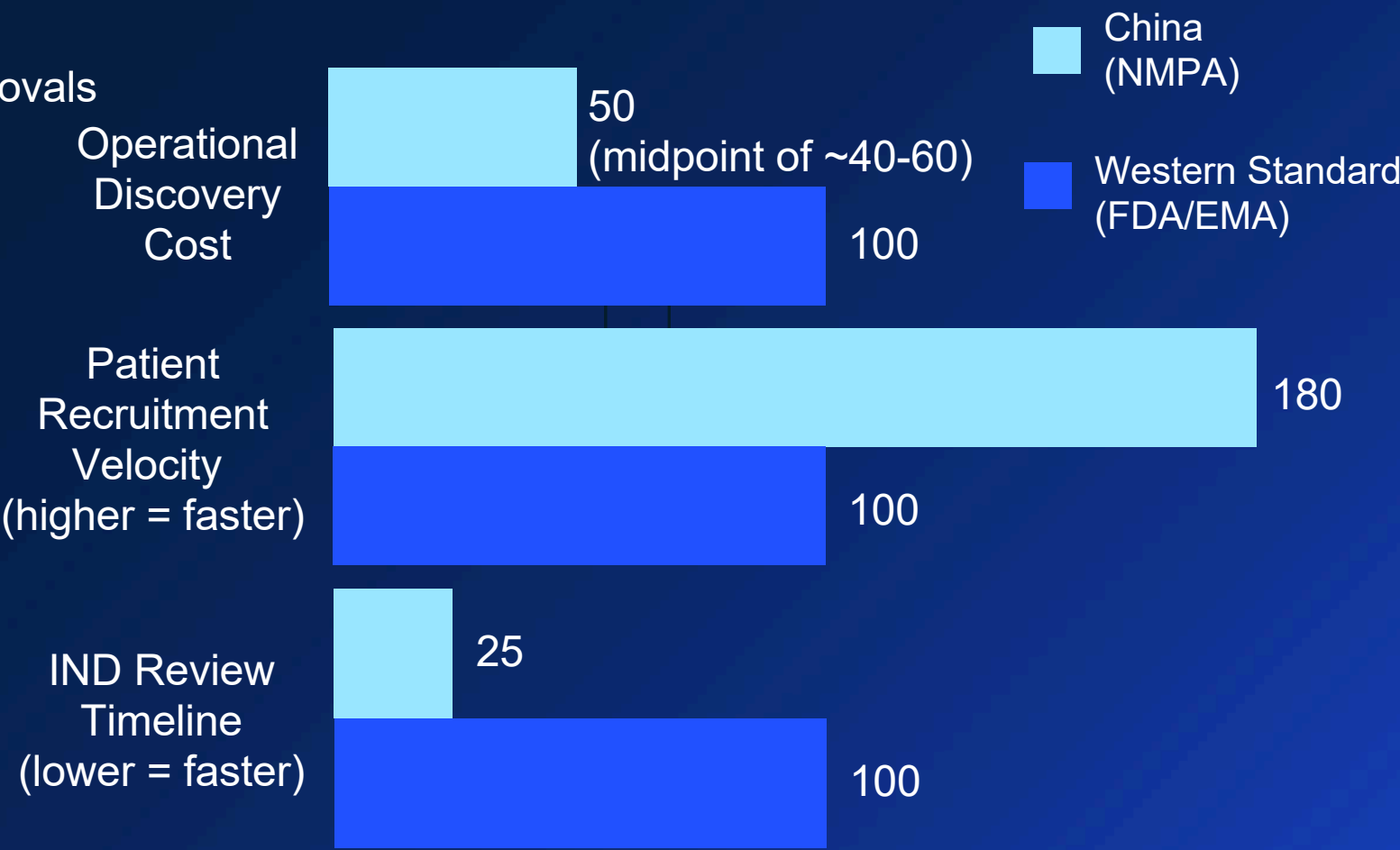
Surging local drug pipelines (NMPA: 83 novel drugs excluding TCM in 2024)



Headwinds

- 1. Standardization & reproducibility still maturing vs. Western benchmarks
- 2. NAM qualification remains less explicit than the FDA’s current roadmap / IStand-style framework
- 3. IP and data-governance concerns for cross-border investors
- 4. Fragmented funding, fewer later-stage scale-up example

The Cost & Validation Speed Advantage (Index)



Appendix

Executive Summary

- Towards Healthcare. Organoids and Spheroids Market Size / Forecast. Reports ~\$1.22B in 2025 and 23.13% CAGR.
- Polaris Market Research. Organ-on-Chip Market Size, Share & Forecast Report 2034. Reports \$163.75M in 2024, \$3.26B by 2034, 34.9% CAGR.
- Global Market Insights. Organoids and Spheroids Market. Reports \$1.8B in 2025, \$9.6B by 2034, 20.3% CAGR.
- Global Market Insights. Organs-on-Chips Market. Reports \$201.5M in 2025, \$2.1B by 2034, 29.7% CAGR.
- FDA. Roadmap to Reducing Animal Testing in Preclinical Safety Studies. Apr. 2025.
- FDA. “FDA Announces Plan to Phase Out Animal Testing Requirement for Monoclonal Antibodies and Other Drugs.” Apr. 10, 2025.
- FDA. “New Approach Methodologies.” Updated 2026.
- DeepTech. Frontier Technologies Empowering New Drug R&D: AI, Synthetic Biology, Gene Delivery, Organoids and Organ-on-Chip. 2023.
- BioWorld. “South Korea forms new organoid consortium for nonanimal testing.” Aug. 2025.

Definition & Scope

- DeepTech. Frontier Technologies Empowering New Drug R&D: AI, Synthetic Biology, Gene Delivery, Organoids and Organ-on-Chip. 2023.
- eMedClub News. Organoid Technology and Application Development White Paper 2023–2024.
- Li et al. “Synergistic innovation in organ-on-a-chip and organoid technologies.” Protein & Cell, 2025.
- Zushin, Mukherjee & Wu. “FDA Modernization Act 2.0: transitioning beyond animal models with human cells, organoids, and AI/ML-based approaches.” Journal of Clinical Investigation, 2023.
- National Academies / NIH. Microphysiological Systems: A Workshop on Barriers, Challenges, and Opportunities. 2021.
- Baran et al. “Perspectives on the evaluation and adoption of complex in vitro models in drug development.” ALTEX, 2022.

Technology Maturity

- Eggshell Research Institute / VBData. Organoid and Organ-on-Chip Industry White Paper, 2023.
- Sino Biological. “Organoid Culture: Definition, Sources, Methods, and Applications.”
- Li et al. “Synergistic innovation in organ-on-a-chip and organoid technologies.” Protein & Cell, 2025.
- Zushin, Mukherjee & Wu. “FDA Modernization Act 2.0: transitioning beyond animal models with human cells, organoids, and AI/ML-based approaches.” Journal of Clinical Investigation, 2023.
- DeepTech. Frontier Technologies Empowering New Drug R&D: AI, Synthetic Biology, Gene Delivery, Organoids and Organ-on-Chip, 2023.
- eMedClub News. Organoid Technology and Application Development White Paper 2023–2024.
- Elveflow. “Organoids vs Organ-on-a-Chip: A Comparative Review.”

Appendix

Global Performance

- VBDData / VCBeat Research & Eggshell Research Institute. Organoid and Organ-on-Chip Industry White Paper, 2023.
- Grand View Research. Organ-on-a-Chip Market Size, Share & Trends Report, 2030. Reports USD157.3M in 2024 and USD952.4M by 2030, 35.11% CAGR.
- McKinsey & Company. “What early-stage investing reveals about biotech innovation.” Dec. 2023.
- Zushin, Mukherjee & Wu. “FDA Modernization Act 2.0: transitioning beyond animal models with human cells, organoids, and AI/ML-based approaches.” Journal of Clinical Investigation, 2023.
- Li et al. “Synergistic innovation in organ-on-a-chip and organoid technologies.” Protein & Cell, 2025.

Geographic Landscape

- Emulate, “Emulate Closes \$82M Series E Financing,” 2021.
- Wyss Institute, “Human Organs-on-Chips.”
- Mimetas, “MIMETAS Secures \$20.5M Series B Financing,” 2018.
- ADICON / EQS News, “ADICON to Acquire Crown Bioscience for US\$204M,” 2025.
- Danaher Life Sciences / Molecular Devices, “Molecular Devices Adds Proprietary Patient-Derived Organoid Technology with Acquisition of Cellesce,” 2022.
- Charles River, “Organoids in Drug Development,” 2024.
- CN Bio, “FDA Further Expands Collaboration with CN Bio,” 2023.
- CN Bio, “CN Bio Raises \$21M Series B First Close,” 2024.
- Market data for CRL and GLW, as of Jun 2026.

Demand Drivers

- FDA. “FDA Announces Plan to Phase Out Animal Testing Requirement for Monoclonal Antibodies and Other Drugs.” Apr. 10, 2025.
- NIH. “NIH to prioritize human-based research technologies.” Apr. 29, 2025.
- FDA. “FDA Achieves Year 1 Goals in Reducing Animal Testing in Drug Development.” Apr. 20, 2026.
- U.S. Congress / Rep. Buddy Carter. “Carter-led bill reducing unnecessary animal testing passes Senate.” Dec. 17, 2025.
- UK DSIT / Gov.uk. “Animal testing to be phased out faster as UK unveils roadmap for alternative methods.” Nov. 11, 2025.
- Zushin, Mukherjee & Wu. “FDA Modernization Act 2.0: transitioning beyond animal models with human cells, organoids, and AI/ML-based approaches.” Journal of Clinical Investigation, 2023.
- Li et al. “Synergistic innovation in organ-on-a-chip and organoid technologies.” Protein & Cell, 2025.
- DeepTech. Frontier Technologies Empowering New Drug R&D: AI, Synthetic Biology, Gene Delivery, Organoids and Organ-on-Chip, 2023.
- eMedClub News. Organoid Technology and Application Development White Paper 2023–2024.
- BIO, Informa PharmaIntelligence & QLS Advisors. Clinical Development Success Rates and Contributing Factors 2011–2020, 2021.

Appendix

Emulate

- Wyss Institute. “Human Organs-on-Chips.”
- Emulate. “Emulate Closes \$82 Million Series E Financing.” 2021.
- Ewart et al. “Performance assessment and economic analysis of a human Liver-Chip for predictive toxicology.” Communications Medicine, 2022.
- FDA. “FDA’s IStand Pilot Program accepts submission of first organ-on-chip technology designed to predict human DILI.” 2024.
- Emulate. “Emulate Launches AVA Emulation System.” 2025.
- NASA. “Artemis II AVATAR.”
- Emulate. “Liver-Chip R1 BioKit Data Sheet.”

Organoid Sciences

- Organoid Sciences. Company website / About Us / History.
- Organoid Sciences. Company website / ODISEI and revenue target disclosure.
- Organoid Sciences. Stock information page, ticker 476040.
- Korea Biomedical Review. “Organoid Sciences targets animal-free drug testing with ODISEI platform ahead of IPO.” Mar. 2025.
- BioWorld. “Three Korea biotechs prep for May IPOs.” May 2025.
- Biz Chosun. “Organoid Science surges over 30% on KOSDAQ debut.” May 2025.

Daxiang

- Daxiang Bio. Product page: IBAC-O1 dynamic co-culture organoid chip.
- Daxiang Bio. Company news / strategic cooperation with Zhongzhi Medical Valley Institute; company profile and platform claims.
- Pedaily. “国产器官芯片企业大橡科技获数千万元A轮融资，鼎晖VGC领投。” May 2021.
- PharmCube / ByDrug. “大橡科技类器官芯片平台屡获资本青睐。” Jul. 2021.
- Jiemian. “大橡科技与尼康签署合作推动类器官芯片技术发展。” Apr. 2023.
- Daxiang Bio. Company news: Chengdu Fifth People’s Hospital strategic cooperation.

Appendix

Potential Opportunity – Hong Kong

- HKU Research. “HKU’s World-First ‘Liver-in-Cube’ Revolutionises Liver Cancer Treatment with Precision.” Published Jan 2026.
- HKUMed. “HKUMed’s world-first ‘Liver-in-Cube’ wins a Gold medal at the 49th International Exhibition of Inventions Geneva.” Published Jun 2024.
- HKU Press Release. “HKU Entrepreneurship Engine Fund × Gobi Partners: Joint fund first close and first batch of investment ceremony.” Published Mar 2026.
- HKIC. “Gobi–HKU Fund I launch / strategic partnership backing HKU technology spin-offs.” Published Mar 2026.
- The Lancet Commission / EurekAlert summary. “Three in five liver cancer cases due to preventable risk factors; cases projected from 870,000 in 2022 to 1.52 million in 2050.” Published Jul 2025.
- The Standard. HKU Liver-in-Cube personalised drug-testing report, including awards recognition. Published Feb 2026.

The Asian Opportunity

- DeepTech. Frontier Technologies Empowering New Drug R&D: AI, Synthetic Biology, Gene Delivery, Organoids and Organ-on-Chip. 2023.
- DIA Global Forum / PharmCube. “China’s New Drug Approvals Again Hit Record High in 2024.” 2025.
- NMPA. “Measures facilitate approval of 48 first-in-class innovative drugs.” 2025.
- FDA CDER. Advancing Health Through Innovation: New Drug Therapy Approvals 2024.
- FDA CDER. Novel Drug Approvals pages, 2020–2024.
- NMPA. Provisions for Drug Registration.
- NIH ClinRegs. China clinical research regulatory profile.
- GlobalData / Clinical Trials Arena. “China’s clinical trial infrastructure delivers up to 60% cost savings.” 2026.
- eMedClub News. Organoid Technology and Application Development White Paper 2023–2024.
- Li et al. “Synergistic innovation in organ-on-a-chip and organoid technologies.” Protein & Cell, 2025.